

Data Protection Impact Assessment (DPIA)

Contents

1. Project Overview	1
1.1 Purpose of the Project	1
2. Data Description and Scope	1
2.1 Types of Data Used	1
2.2 Data Subjects	1
2.3 Data Volume and Frequency.....	1
3. Lawful Basis for Processing	2
3.1 Legal Basis under UK GDPR	2
4. Data Flow and Access Controls	2
4.1 Data Flow Description	2
4.2 Data Storage and Retention.....	2
4.3 Access and Security Controls	2
5. Risk Assessment	2
6. Data Sharing and Third Parties.....	3
7. Data Subject Rights	3
8. Sign-Off and Approval	3
9. Review Schedule	3

1. Project Overview

1.1 Purpose of the Project

NHFT delivers over 500 services across community, mental health, and specialist pathways. Each service manages fluctuating demand and capacity, often relying on manual spreadsheets and retrospective analysis. This project will develop a generic, configurable Demand and Capacity Modelling Platform that enables accurate forecasting, simulation, and optimisation of service activity.

The system will provide:

- Forecasts of service demand using statistical and machine learning models.
- Capacity modelling to estimate resource needs (staffing, sessions, rooms).
- Simulation of different operational scenarios.
- A secure, internal dashboard for service managers to interact with and interpret outputs.

The project is designed to improve resource utilisation, reduce breaches, and enhance patient access, contributing to NHFT's operational efficiency and service quality.

2. Data Description and Scope

2.1 Types of Data Used

The project will process pseudonymised or aggregated operational data from NHFT's existing BI systems.

Dataset	Fields / Variables	Sensitivity	Source
Referrals	Referral date, service ID, referral source, urgency	Low–Medium	ESR/BI Warehouse
Appointments	Appointment date, duration, outcome, DNA flag	Low–Medium	BI Warehouse
Workforce	Staff grade, WTE, sessions, availability	Medium	ESR
Service Metadata	Service name, site, type (inpatient/outpatient/community)	Low	BI Master Data
Calendar	Holiday/term indicators	Low	External public dataset

No direct identifiers (e.g., name, NHS number, address, DOB) will be accessed. All records will be pseudonymised at source.

2.2 Data Subjects

- Patients receiving NHFT services (aggregated and pseudonymised).
- Staff (aggregated, e.g., total WTE, capacity metrics).

2.3 Data Volume and Frequency

Data will be accessed via secure SQL queries from existing BI systems. No manual data extracts or external data storage will be used. The process will be automated via direct queries, ensuring that data are not retained locally or duplicated.

3. Lawful Basis for Processing

3.1 Legal Basis under UK GDPR

The project processes data under the lawful basis of:

- **Article 6(1)(e):** Processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller (NHS Trust).
- **Article 9(2)(h):** Processing is necessary for the management of health or social care systems and services.

The project does not require explicit consent as it uses existing operational data within the boundaries of NHFT's public task and Information Governance framework.

4. Data Flow and Access Controls

4.1 Data Flow Description

1. Data extracted securely via SQL from NHFT's BI data warehouse.
2. Data transformed and pseudonymised automatically within the NHFT environment.
3. Processed data loaded dynamically into the modelling framework for analysis.
4. Model outputs (aggregated forecasts, utilisation metrics) displayed in a secure internal dashboard.
5. No personal-level data exported or shared externally.

4.2 Data Storage and Retention

- Data are not stored locally; models operate through **automated, read-only queries**.
- No raw datasets are retained beyond processing runtime.
- Outputs are aggregated and retained within NHFT's secure Power BI or internal servers for reporting purposes.

4.3 Access and Security Controls

- Access limited to authorised BI staff with IG and Data Security training.
- Role-based access controls (RBAC) enforced.
- Encrypted connections (SSL) and NHFT network firewall protection.
- System activity and query logs retained for audit.

5. Risk Assessment

Risk ID	Description of Risk	Likelihood	Impact	Mitigation / Safeguard
R1	Unauthorised access to pseudonymised data.	Low	High	Access restricted to authorised users. Encrypted network and MFA authentication.
R2	Potential re-identification of individuals from small datasets.	Low	Medium	Aggregation thresholds and suppression rules to limit re-identification of small counts (<5).
R3	Model bias due to incomplete or unbalanced data across services.	Medium	Medium	Regular fairness evaluation and model parameter tuning to maintain equity.

R4	Data extraction or transformation error leading to inaccurate outputs.	Medium	Medium	Automated data validation checks and version-controlled pipelines.
R5	Breach of open-source software licence.	Low	Medium	Use only well-documented libraries under permissive licences (MIT/BSD).
R6	Misinterpretation of model outputs by non-technical users.	Medium	Medium	Include documentation, tooltips, and confidence intervals in dashboards.
R7	Retention of temporary data files outside of secure environment.	Low	High	Implement direct-query automation to remove need for intermediate storage.

6. Data Sharing and Third Parties

- No external sharing of identifiable or pseudonymised data.
- Aggregated outputs may be presented to internal NHFT stakeholders.
- The University of Nottingham may view non-disclosive summaries for assessment purposes under the apprenticeship agreement.
- Any future data sharing will require a formal Data Sharing Agreement (DSA) approved by NHFT Information Governance.

7. Data Subject Rights

Although data are pseudonymised and not processed at the identifiable level, data subjects retain their general rights under UK GDPR, including:

- Right to be informed (covered by NHFT privacy notices).
- Right of access (via NHFT Subject Access Request procedures).
- Right to restrict or object to processing (where applicable).
- Right to erasure (not applicable to pseudonymised operational data).

8. Sign-Off and Approval

Name	Role	Signature	Date
Yahya Hafeji	Project Lead (BI Developer)	YH	04/11/2025
Ed Buxton	BI Manager (Information Asset Owner)	EB	08/12/2025
Adam Shelley	Information Governance Representative	AS	05/12/2025

9. Review Schedule

- **Initial DPIA Approval:** 08/12/2025
- **Review Interval:** Upon significant change in project scope, data source, or model design.